

Engagement Prediction for Non-Speaking Participants in Online Meetings

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Online meetings have become a common mode of collaboration, yet automatically estimating engagement remains challenging when participants are not speaking. While engagement prediction has focused on speakers using prosodic-acoustic cues[1], non-speaking participants primarily express engagement through embodied, nonverbal behavior. This paper investigates whether multimodal nonverbal features, together with behavioral alignment from co-present participants, can improve engagement prediction for non-speaking participants under semi-supervised learning methods.

We formulate engagement prediction as a semi-supervised clustering-based classification problem with three engagement states: Disengagement, Low engagement, and Active engagement. Experiments are conducted on a subset of the GEHM Zoom meeting corpus[2]. Participant videos are segmented into overlapping 5-second clips to capture alignment patterns while minimizing the temporal noise of extended windows. Segments containing speech are excluded. For each segment in which a participant is not speaking, multimodal features from that participant are automatically extracted, including head pose and head movement statistics, nod and shake events, hand gestures, facial action unit (AU) features, and deep visual representations obtained using CLIP[3]. Dimensionality reduction is applied to high-dimensional AU and CLIP features using PCA. Given the intuition that attentive participants listening to the same content tend to exhibit similar and synchronized nonverbal behavioral patterns, we introduce alignment features that quantify behavioral similarity between the target participant and two co-present non-speaking participants from the same meeting time frame. These features include cosine similarity computed separately on PCA-reduced AU features and PCA-reduced CLIP embeddings, capturing facial expression-level and global appearance-level alignment, respectively.

To estimate engagement values, we compare three clustering algorithms: K-Means, Gaussian Mixture Models (GMM), and Hierarchical clustering, using two feature configurations: target-only features and target features augmented with alignment features. Cluster labels are assigned using a minimal supervision strategy in which a small number of samples per cluster are annotated manually, and the majority label is propagated to all samples within the corresponding cluster. Performance is evaluated on a held-out, human-annotated test set using weighted F1-score. See Table 1 for figures on annotation agreement.

Results show that incorporating alignment features improves engagement prediction performance for K-Means clustering, increasing the weighted F1-score from 0.31 to 0.40, with a substantial improvement for Active engagement (F1 0.32 $\rightarrow$ 0.63). In contrast, GMM and Hierarchical clustering exhibit limited performance and often collapse to a single dominant class, regardless of feature configuration. These findings indicate that nonverbal multimodal cues provide useful signals for engagement prediction, and that behavioral alignment patterns offer additional discriminative information when combined with appropriate clustering methods. The study highlights both the potential and the limitations of semi-supervised clustering approaches for modelling engagement in non-speaking participants during online meetings.

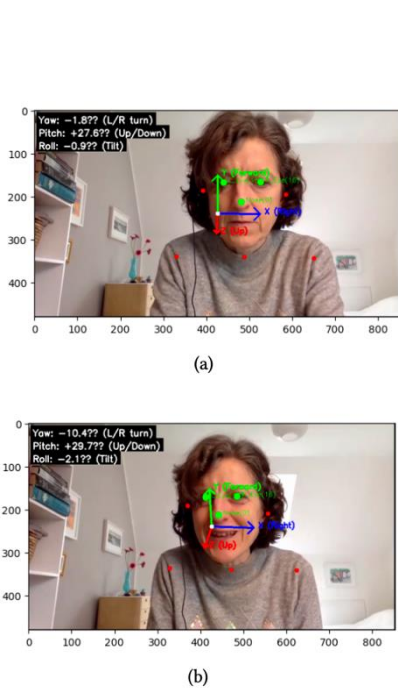


Figure 1: Facial feature extraction and 3D head-pose estimation. The system utilizes a coordinate axis to calculate Yaw, Pitch, and Roll, while keypoints identify critical landmarks including the eyes, chin, and shoulder alignment for spatial tracking.

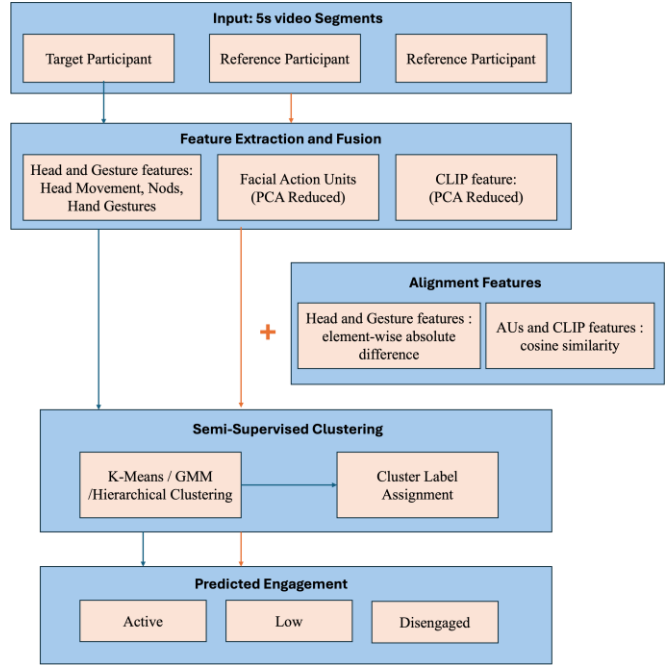


Figure 2: Multimodal semi-supervised framework for non-speaking engagement prediction. The pipeline extracts individual and behavioral alignment features from 5-second intervals. These features are processed through clustering algorithms to predict participant engagement levels based on minimal supervision.

Class	Precision	Recall	F1-score	Fleiss' Kappa
Active Engagement	0.51	0.82	0.63	0.45
Low engagement	0.33	0.18	0.23	0.22
Disengagement	0.33	0.06	0.10	0.53
Weighted avg	0.42	0.47	0.40	

Table 1: Classification performance and label reliability analysis. Performance remains limited for Low engagement ($F1\ 0.30 \rightarrow 0.23$) and Disengagement ($F1\ 0.15 \rightarrow 0.10$). Low agreement for the "Low engagement" label ($\kappa = 0.22$) suggests inherent ambiguity, while the Disengagement class (19% of data) likely suffers from minority status and weak cluster coherence.

References

- [1] A. Pellet-Rostaing, R. Bertrand, A. Boudin, S. Rauzy, P. Blache, A multimodal approach for modeling engagement in conversation, *Frontiers in Computer Science* 5, 2023, 1062342. doi:10.3389/fcomp.2023.1062342.
- [2] P. Paggio, M. Agirrezabal, C. Navarretta, L. Vitasovic, Multimodal behaviour in an online environment: The gehm zoom corpus collection, in: *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, ELRA and ICCL, 2024, pp. 11890–11900.
- [3] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning transferable visual models from natural language supervision, in: *Proceedings of the 38th International Conference on Machine Learning*, PMLR, 2021, pp. 8748–8763. doi: 10.48550/arXiv.2103.00020.